📊 Data Science Roadmap (Week-wise Structured Plan)

🧠 PHASE 1: Foundations (Python + Basics)

📅 Weeks 1–6

📘 Week 1: Introduction to Data Science

- What is Data Science?
- Roles:
  - Data Analyst
  - Data Scientist
  - Data Engineer

👉 Setup:

- Install Python
- Use Jupyter Notebook / VS Code

📘 Week 2: Python Basics

- Variables
- Data types
- Lists, tuples, dictionaries

👉 Practice:

- Basic Python exercises

📘 Week 3: Control Flow

- if/else
- loops

👉 Practice:

- Logic problems

📓 Week 4: Functions & Modules

- Functions
- Importing libraries

📓 Week 5: NumPy

- Arrays
- Mathematical operations

👉 Practice:

- Matrix operations

📓 Week 6: Pandas Basics

- DataFrames
- Reading CSV files

👉 Practice:

- Load and explore datasets

🧠 PHASE 2: Data Analysis & Visualization

📅 Weeks 7–14

📓 Week 7: Data Cleaning

- Handle missing values
- Remove duplicates

📓 Week 8: Data Manipulation

- Filtering
- Grouping

📓 Week 9: Data Visualization Basics

- Matplotlib

📓 Week 10: Advanced Visualization

- Seaborn

👉 Practice:

- Plot graphs (bar, line, heatmap)

📓 Week 11: Exploratory Data Analysis (EDA)

- Understand patterns
- Detect trends

📓 Week 12: Statistics Basics

- Mean, median, mode
- Variance, standard deviation

📓 Week 13: Probability

- Basics
- Distributions

📓 Week 14: Mini Project

👉 Perform:

- Full EDA on dataset

🟢 PHASE 3: SQL + Data Handling

📅 Weeks 15–18

📓 Week 15: SQL Basics

- SELECT
- WHERE
- ORDER BY

📓 Week 16: SQL Advanced

- JOINs
- GROUP BY

📓 Week 17: Working with Databases

- Connect Python with SQL

📓 Week 18: Project

👉 Build:

- Data analysis using SQL + Python

🧠 PHASE 4: Machine Learning

📅 Weeks 19–28

📘 Week 19: Intro to Machine Learning

- Types:
  - Supervised
  - Unsupervised

📘 Week 20: Linear Regression

- Concepts
- Implementation

📘 Week 21: Logistic Regression

- Classification basics

📘 Week 22: Evaluation Metrics

- Accuracy
- Precision / Recall

📘 Week 23: Decision Trees

- Tree models

📘 Week 24: Random Forest

- Ensemble learning

📓 Week 25: Clustering

- K-Means

📓 Week 26: Feature Engineering

- Feature selection

📓 Week 27: Model Optimization

- Hyperparameter tuning

📓 Week 28: ML Project

👉 Build:

- Prediction model (real dataset)

🧠 PHASE 5: Advanced Tools & Visualization

📅 Weeks 29–36

📓 Week 29–30: Tableau / Power BI

- Dashboards
- Reports

📓 Week 31–32: Advanced Visualization

- Storytelling with data

Week 33–34: Big Data Intro

- What is Big Data?
- Tools:
  - Hadoop
  - Spark

Week 35–36: Deployment Basics

- Flask / Streamlit
- Deploy ML model

PHASE 6: Projects (Portfolio Building)

Weeks 37–44

Week 37–38: Beginner Projects

- Sales analysis
- Data cleaning project

Week 39–40: Intermediate Projects

- Dashboard (Power BI/Tableau)
- EDA project

Week 41–42: Advanced Projects

- Machine learning project
- Prediction system

📓 Week 43–44: Final Project

👉 Build:

- End-to-end pipeline:
  - Data → Model → Dashboard

🧠 PHASE 7: Job Preparation

📅 Weeks 45–52

📓 Week 45: Resume Building

- Highlight:
  - Projects
  - Tools

📓 Week 46: Portfolio

- GitHub
- Kaggle profile

📓 Week 47: Interview Prep

- SQL questions
- ML basics

📓 Week 48: Case Studies

- Business problems

📓 Week 49–50: Practice

- Daily problems
- Mock interviews

📓 Week 51–52: Apply for Jobs

- Internships
- Junior Data Analyst roles

🔄 Weekly Routine

- 5 days learning
- 2 days project/practice

Daily:

- 2–4 hours

🎯 Final Outcome (After 1 Year)

By Week 52:

✅ Strong Python + Pandas
✅ Data visualization skills
✅ SQL mastery
✅ Machine learning basics
✅ 4–5 strong projects
✅ Portfolio (GitHub + dashboards)

👉 Job Roles:

- Data Analyst
- Junior Data Scientist
- BI Analyst